$\omega = 2\pi f$

$$\frac{\theta_b}{\theta_s} = P(s) = \frac{K}{(s-p_1)(s-p_2)}$$

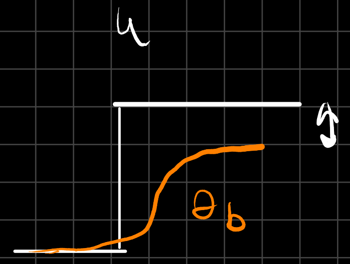
$K = 138{,}3035$

$p_1 = -13{,}2515 \text{ rad/s} \longrightarrow -2{,}1090 \text{ Hz}$

$p_2 = -32{,}6059 \text{ rad/s} \longrightarrow -5{,}1894 \text{ Hz}$

$$x = \begin{pmatrix} \theta_b \\ \dot{\theta}_b \end{pmatrix} \rightarrow \dot{x} = A x + B . u \longrightarrow \text{ángulo del servo}$$

$$y = C \cdot x + D . u \quad (D = 0)$$



$C = (1 \quad 0)$

$$\theta_b (s-p_1)(s-p_2) = K \cdot \theta_s$$

$$\left.\frac{\theta_b}{\theta_s}\right|_{s=0} = \frac{K}{p_1 \cdot p_2} = \text{Ganancia en continua} = \frac{\theta_{b}\text{-final}}{\theta_s\text{-final}}$$

$$\theta b(s)(s^2 - (p_1+p_2).s + p_1 . p_2) = K \cdot \theta_s(s)$$

$$\mathcal{L}^{-1}: \ \ddot{\theta}_b - (p_1+p_2)\underbrace{\dot{\theta}_b}_{x_2} + p_1 p_2 . \underbrace{\theta_b}_{x_1} = K . \theta_s$$

$$K = \frac{\theta_{bf}}{\theta_{sf}} \cdot p_1 . p_2$$

$$\dot{x}_2 = -p_1 p_2 x_1 + (p_1 + p_2) x_2 + k u$$

$$\begin{cases} \dot{x}_1 = x_2 \\ \dot{x}_2 = -p_1 p_2 x_1 + (p_1 + p_2) x_2 + k \cdot u \end{cases}$$

$$\Rightarrow \dot{x} = \begin{pmatrix} 0 & 1 \\ -p_1 p_2 & p_1 + p_2 \end{pmatrix} x + \begin{pmatrix} 0 \\ k \end{pmatrix} \cdot u$$

$$y = c \cdot x$$

De matlab temenos $L = \begin{pmatrix} -0{,}0039 \\ 1{,}3612 \end{pmatrix}$

$$\hat{x}_k = A_d \hat{x}_{k-1} + B_d \cdot u_{k-1} + L(y_{k-1} - C_d \cdot \hat{x}_{k-1})$$

Ad =

1.0000	0.0200
-8.6415	0.0829

Bd =

0
2.7781

Cd =

1	0